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S.300 EARTH WORKS

1. EXCAVATION

1.1 Scope of Work

Excavation shall include incidental clearing, grubbing, grading, compacting, levelling, excavation and disposal of excavated materials, protection and support required, sheeting, shoring, bracing and cofferdams, dewatering of excavation and work areas, and preparation of bearing areas as required to properly install and complete the work, regardless of portion of the work for which dewatering is required and regardless of the nature of materials encountered in excavation, including construction of temporary diversion ditches, culverts, temporary access road etc., where required.

1.2 Execution of Excavation

- 1.2.1. The Bidder shall notify the Owner in writing at least three (3) working days, prior to the start of any excavation, the beginning of the work in any working area to enable the Owner to verify all necessary levels and cross sections of the original ground surface. The Bidder shall also, prior to starting excavation in any area, submit for the Owner's approval a plan of operation with complete details of the method of excavation, type of equipment to be used, proposed method of diversion of drainage, safety measures to be applied, etc.
- 1.2.2. The Bidder shall set out all works as shown on the Bidder's drawings, utilizing the bench marks and coordinates furnished by the Owner.
- 1.2.3. The Bidder shall be responsible for the correct setting out of work. Any additional work caused by his negligence in setting out shall be at his expenses and corrective action must be taken without delay as directed by the Owner.
- 1.2.4. The Bidder shall be responsible for all necessary safety measures from the commencement of work to the completion. The Bidder shall strictly follow safety regulations in order to prevent accidents.

- 1.2.5. The Bidder must take every precaution and use the most appropriate methods of excavation and equipment to avoid the loosening of material or the breaking of rock beyond the lines and grades shown on the Bidder's drawings or as directed by the Owner. If for any reason whatsoever excavation is carried out beyond the lines and grades, the Bidder shall, at his own expense, take the necessary measures to restore lines and grades with approved materials such as backfill or concrete and in such a way as may be directed by the Owner.
- 1.2.6. Bottoms of all excavations must be trimmed to line and grade, this being subject to the Owner's approval. When requested by the Owner, the final 20cm of excavation in any loose geological formation where concrete is to be placed shall be excavated by hand tools to avoid disturbance of the bottom.
- 1.2.7. Particular care shall be exercised when excavating in the vicinity of existing and partially or fully completed structures. The Bidder shall repair or replace, at his own expense, any structures or equipment damaged by his actions.
- 1.2.8. It will be the Bidder's responsibility to utilize to the fullest extent all materials from excavations which are approved and deemed suitable by the Owner as backfill or other construction. The Bidder shall make his best efforts to comply with the above requirements.
- 1.2.9. Where possible, suitable materials shall be excavated separately from materials to be wasted and shall be placed in the designated final locations directly from the excavation or shall be stockpiled as directed by the Owner, if the immediate placement in the designated location is not possible.
- 1.2.10. All related permits and approvals from authorities shall be obtained by the Bidder as per local regulations.
- 1.2.11. Excavation to indicated elevations and dimensions shall be within a tolerance of plus or minus 25 mm (1 inch) as per AASHO.

1.3 Protection and Support

- 1.3.1. The Bidder shall provide protection and support required to properly install work for protection and support of adjacent structures and improvements as required for the safety of traffic and life .
- 1.3.2. Protection and support shall include temporary sheeting, bracing and shoring and where indicated, permanent sheeting, bracing and shoring. Temporary sheeting, bracing and shoring shall be removed by the Bidder when their use is no longer required, unless otherwise directed by the Owner.
- 1.3.3. Excavation slopes shall be protected and supported, where required or when directed by the Owner, so that the slopes and bottoms will be maintained for depth, width and condition indicated and so that adjacent structures or other construction will be protected from damage caused by any earth or rock movement.
- 1.3.4. Protection and support shall be arranged for minimum interference with pipe laying, electrical installation and other similar work.
- 1.3.5. The Bidder shall submit detailed calculations and drawings for the prior approval of the Owner, however such approval shall not relieve him from full responsibility for the adequacy of the temporary supports.

1.4 Dewatering and Diversion of Water

1.4.1 General

- 1.4.1.1. The Bidder shall furnish, install and operate necessary incidental dewatering equipment as required to do all excavation in dry, to the satisfaction of the Owner.
- 1.4.1.2. Except where specifically permitted by the Owner, no surface runoff shall be ponded or restricted to a greater degree than would have occurred naturally either before the beginning of construction or after completion of the work.

1.4.2 Diversion of Surface Runoff

- 1.4.2.1. During the excavation and backfill work, the Bidder shall construct temporary drainage ditches, culverts and other structures as required and as directed by the Owner to divert/drain the work areas.
- 1.4.2.2. Prior to start of the work, the Bidder shall submit to the Owner for approval drawings and data indicating his proposed plans for diversion and care of surface runoff.
- 1.4.2.3. The Bidder shall not place any fill across routes of natural drainage channels until the Bidder has made provisions to drain the surface runoff into a new diversion ditch.
- 1.4.2.4. No surface runoff shall be ponded or restricted to a greater degree than would have occurred naturally either before the beginning of construction or after construction of the work, unless approved by the Owner. Any damage to any property, equipment as the result of noncompliance to the above shall be sole responsibility of the Bidder.

1.4.3 Dewatering

- 1.4.3.1. The excavation and the reinforced concrete work included in this Contract shall require installation of a dewatering system consisting of cofferdam/cutoff walls, sumps and pumps of adequate capacities, sealing of cracks or fissures in the rock and other associated works. The Bidder shall design, furnish and install a complete dewatering system and maintain the dewatering system in operating condition until the completion of this work.
- 1.4.3.2. The Bidder shall submit to the Owner for approval a preliminary design for the dewatering system proposed to be installed by him prior to start of excavation work.
- 1.4.3.3. The Bidder shall be responsible for the safety and functioning of the dewatering system.

- 1.4.3.4. The pumps, piping and the sumps installed shall have adequate capacity to keep the excavation dry for the entire construction period.
- 1.4.3.5. If any failure of the dewatering system occurs, the Bidder shall be responsible for correction of all deficiencies in time to the satisfaction of the Owner.
- 1.4.3.6. The Bidder shall make available a standby electric supply system, or spare diesel pumps and spare parts for quick replacement for the continuous service of the dewatering system.

1.5 Disposal of Excavated Materials

- 1.5.1. Excavated materials which are not suitable as fill materials shall be hauled and transported to and disposed of into spoil area off the Project Site proposed by the Bidder and approved by the Owner. The materials which are in excess of the fill requirement for the Bidder's work shall be hauled to the designated stockpile area. Before transporting to the stockpile area, the Bidder may be required to segregate and screen large size stones to be placed in separate stockpile areas, however Bidder shall be responsible for land acquisition of disposal area..
- 1.5.2. The spoil area and the stockpile area shall be leveled, trimmed and graded to drain as directed by the Owner.

1.6 Earth Excavation

- 1.6.1. Earth excavation means general excavation, including excavation from project areas, of material such as clay, silt, sand, gravel, crushed or screened rocks, boulders, cemented or weathered rock which may be removed by hand tools, power shovels, dozers or other earth-moving equipment and by ripping without continuous systematic drilling and blasting. Rippable material shall be defined as such that the materials can be removed by a properly equipment.
- 1.6.2. Earth excavation may be accomplished by any method and by the use of any excavation and hauling equipment suitable for the work. Rock or boulders encountered in the excavation may be drilled and blasted whenever other

methods of loosening such as ripping, crowbarring, or wedging prove impracticable and shall be removed as required.

- 1.6.3. In case that earth excavation shall not be carried out to rock but to specified levels as indicated on the Bidder's drawings and the natural foundation material is disturbed or loosened for any reason, the foundation shall be improved by compactson, or replaced with approved fill and compacted, or by other means as directed by the Owner.

1.7 Trench Excavation

- 1.7.1. Trenches shall be to the limit lines and grades shown on the Bidder's drawings or as directed by the Owner.
- 1.7.2. The Bidder shall furnish all labor, material and equipment required for the proper execution of the work. Correct and adequate bracing and supporting of the excavation shall also be allowed for in his own cost.
- 1.7.3. Excavation performed beyond the lines and grades shown on the Bidder's drawings, without specific instruction from the Owner, shall be backfilled to grade and lines with approved materials. All additional work of this nature shall be carried out at the Bidder's expense.
- 1.7.4. Trenches for pipelines, steel pipes, ducts and conduits shall be excavated in straight sections and provide a minimum of 20cm between the outside face of the pipelines, electrical ducts or conduit and the side of the trench.
- 1.7.5. Mechanical excavation shall stop not less than 5cm above the final bottom level and also at an approved distance from permanent installations so that no damage occurs to the same. The remainder of the excavation shall be done and graded manually to provide uniform bearing on compacted soil immediately before the installation or placing of any equipment. No additional payment will be made for such manual excavation and the cost thereof shall be included in the Contract Price.
- 1.7.6. If the character of the material at any location in the bottom of a pipe trench is such that it might cause unequal settlement or provide unequal bearing for the pipe, or is otherwise unsuitable for the pipe's foundation, unsatisfactory

materials shall be removed to such a depth as may be directed by the Owner and backfilled with granular materials and/or concrete and compacted as directed by the Owner without additional cost to the Owner.

1.8 Excavation for the Bidder's Facilities Requirement

1.8.1. This classification shall consist of all excavation performed by the Bidder for the purpose of executing the work, but not specifically shown, required or mentioned in the Contract, such as excavation for haul roads, camps, shops, sanitary, land fills and other similar excavation. Permission shall be obtained from the Owner for the location of borrow areas and eventual spoil areas for unsuitable material.

1.8.2. The Bidder shall stabilize the slopes of such excavation in such a manner that they do not represent any danger to the public or property. Such excavation slope stabilization shall be performed at the Bidder's own cost and no separate payment will be made for this work.

1.9 Preparation of Bearing Areas

1.9.1. Bearing areas for foundations shall be inspected and approved by the Owner before any concrete is placed thereon. If bearing areas are not suitable, as determined by the Owner, the Bidder may be requested to carry the excavation deeper to more suitable bearing material or to improve bearing capacity without additional cost. The Bidder may be requested by the Owner to make auger boring or other tests in bearing areas to determine thickness of bearing stratum.

1.9.2. Foundations shall be placed on undisturbed soil unless otherwise indicated.

1.9.3. Before placing any concrete for beams or slabs on fill, the bearing soil shall be compacted as specified herein for type of fill material involved. Surfaces shall be relatively smooth.

1.9.4. The surface of the soil shall be thoroughly wetted with clean water immediately before sand bedding and lean concrete is placed.

2. FILL

2.1 Scope of Work

2.1.1. The scope of work under this section covers excavation of materials from the stockpile area if required, hauling to the places of fill, spreading, compacting, grading, finishing, and incidental dewatering where required for the general site fill, construction of the core for any embankment, structural backfill, trench backfill and other miscellaneous fill.

2.1.2. The Bidder shall perform filling/backfilling work in accordance with this Specification and as shown the Bidder's drawings or as directed by the owner.

2.2 Material Sources

The sources of material shall be materials previously excavated at the site by the Bidder or from outside sources. The fill material shall be approved by the Owner prior to placement.

2.3 Fill/Backfill Placement

2.3.1 Compacting Equipment

2.3.1.1. Compacting equipment shall be of such design, weight and quality as is required to obtain the densities specified herein.

2.3.1.2. Areas inaccessible to self-propelled compacting equipment shall be compacted by hand operated mechanical tampers or vibrators.

2.3.1.3. Self-propelled compacting equipment, i.e., sheep's foot rollers and segmental pad rollers, and hand operated mechanical tampers and vibrators, shall efficiently perform compaction in accordance with this Specification and as indicated.

2.3.1.4. The Bidder's plea that the equipment he has available at the Project Site cannot produce the densities specified will not be accepted by the Owner. The Bidder shall furnish equipment as required.

2.3.2 Structural Fill Material

2.3.2.1 Preparation of Sub-grade

Sub-grade to receive structural fill material will be inspected by the Owner to determine if it is suitable and has sufficient bearing capacity for the material and loads to be placed on it.

If sub-grade is not suitable, as determined by the Owner, the Bidder may be directed to perform additional excavation with compensation thereof.

2.3.2.2 Compaction Densities

Material shall be brought up to grade elevation indicated on the Bidder's drawings, with suitable moisture control and compaction throughout placement, to produce the following densities:

- a. Type structural fill material: to 90 percent of maximum dry density per ASTM D1557, Test Method indicated
- b. Type SFG material: to a minimum of 85 percent of relative density per ASTM D2049, test Method indicated

2.3.2.3 Placing Material

- a. Unless otherwise indicated, structural fill material shall be placed within two (2) percent of the optimum moisture content as applicable, in uniform layers not over 150mm (loose measure).
- b. The Bidder shall add water or dry out material to maintain the optimum moisture content throughout placing and compacting.

2.3.3 Random Fill Materials

2.3.3.1 Random fill materials shall be used for filling excavated trenches and as site area fill where designated.

2.3.3.2 Materials shall be usable fill materials from the excavation or fill materials from the outside sources as approved by the Owner.

2.3.3.3 Fill in the excavated area shall be placed in layers and method of placement of the fill shall be approved by the Owner. Prior to the start of the fill placement, the Bidder shall submit a statement describing in detail his proposed method of fill placement area for the Owner's approval.

- 2.3.3.4 The fill shall be placed in layers not exceeding thickness of 300 mm (loose measure) and compacted to produce densities to minimum 85 percent relative density per ASTM D2049, Test Method indicated.

2.4 Backfill around Structures

Backfill shall be placed within the limits of the excavations or to the lines and grades shown on the approved Bidder's drawings in 20cm thick horizontal layers and compacted carefully. Backfill shall not be placed until the structure has been inspected and approved by the Owner, concrete has been cured sufficiently to achieve specified compressive strength, the forms removed, the excavation cleaned of trash, debris and water and the concrete has received its bituminous coatings (if directed or shown on the Bidder's drawings). Where backfill is placed on more than one side of a structure, it shall be placed and compacted evenly to maintain similar fill elevations on all sides of the structure at the same time. Granular material used as structure backfill shall be compacted to a minimum of 85 percent of relative density per ASTM D2049, Test Method indicated.

2.5 Backfill of Pipe Trenches and Cuts

2.5.1 General

- 2.5.1.1. Backfill around pipe shall not be performed until pipe runs have been bedded and the Owner has inspected, tested and approved the piping installation.
- 2.5.1.2. Care shall be exercised in placing and compacting material around pipes to maintain the material approximately at the same level, not in excess of a 300mm differential, on both sides of the pipe, throughout the placing and compacting of the material.
- 2.5.1.3. Placing and compacting of material shall be performed simultaneously on both sides of pipes in such a manner that will prevent injurious side pressures on piping.
- 2.5.1.4. Material shall be thoroughly compacted using hand operated power tampers until compacted material is at least 600mm above pipe.

- 2.5.1.5. Self-propelled compaction equipment shall not be used above pipe until the compacted material has reached 600mm minimum, unless approved by the Owner.
- 2.5.1.6. When a flat bed is required for the bottom of a trench for pipes, granular material shall be provided for bedding to depths indicated on the Bidder's drawings. If unsuitable soil, as determined by the Owner, is encountered and is directed to be removed, the Bidder shall provide the additional depth of granular compacted bedding required. Bedding for concrete cuts shall be provided if indicated on the Bidder's drawing.
- 2.5.1.7. **Bedding and Backfilling Material Placement**
- a. For coated pipe, granular bedding and backfill material shall be crushed stone, coarse sand or gravel or other material approved by the Owner. Unless otherwise directed by the Owner, bedding and backfilling material shall conform to ASTM C33, and shall not be larger than 10mm in size.
Bedding material, material under haunches and along sides of pipe of a point 150mm above the top of the pipe shall be this material. Care shall be taken in placing material to prevent damage to pipe coating.
 - b. For uncoated pipe, granular bedding and backfill material shall be type SFG. Bedding material and material under haunches and along sides of pipe to a pipe 150mm above the top of pipes shall be this material.
 - c. Compaction densities for material (both ASTM C33 and Type SFG) shall be as per Sub clause 2.3.3.
 - d. Material from 150mm above the top of pipes to indicate grade shall be selected material approved by the Owner. This material shall be placed and compacted as per Sub clause 2.3.2 and 2.3.3 as applicable for the materials to be used.

2.5.2 Backfilling of Fiberglass Reinforced Pipe

- 2.5.2.1. Plastic FR pipe includes RTRP (Reinforced Thermosetting Resin Pipe), GRPP (Glassfiber Reinforced Plastic Pipe) and RPMP (Reinforced Plastic Motor Pipe) and may be of any type or grade, lined or unlined, as per ASTM D2310.

- 2.5.2.2. The manufacturer of the pipe shall be advised by the Bidder of the soil conditions expected to be encountered in bedding and backfilling pipe trenches or cuts. Promptly thereafter the pipe manufacturer shall submit recommended procedures for cut, slope, bedding material, bedding depth, haunch and fill over pipe utilizing either available material from the Site, or more appropriate material obtained from the offsite. These procedure will be reviewed by the Owner (or the testing laboratory) prior to placement of the glassfiber reinforced pipe.
- 2.5.2.3. Bedding and haunch material (to a minimum height above the top of pipe of 300mm) will be Type Structural Fill material. Material above Type SFG material shall be random fill or as directed by the Owner.
- 2.5.2.4. Depth of each layer of material placed shall be as directed by the pipe manufacturer, but not greater than 150mm (loose fill layer thickness) and shall be compacted using portable tampers.
- 2.5.2.5. Care shall be taken during placement to assure evenness of compaction densities, especially under points between pipe sections, as well as to avoid floatation of the pipe.

2.6 Backfilling of Electrical Duct Banks and Direct Burial Cable

2.6.1 Duct Banks

- 2.6.1.1. Material directly above and on both sides of duct banks may be material excavated from the trench or cut, unless otherwise indicated by the Owner, providing that the maximum size of stone in the material shall not be greater than 50mm.
- 2.6.1.2. Compaction shall conform to the requirements of Subclause 2.3.2 and 2.3.3 as applicable.

2.6.2 Direct Buried Cable

- 2.6.2.1. Material around direct buried cable may be material excavated from the trench or cut, providing it is free from debris or coarse aggregate. Maximum size of stone shall not exceed 50mm in the fill material.
- 2.6.2.2. Compaction shall confirm to the requirements of Subclause 2.3.2 and 2.3.3 as applicable.

2.7 Backfilling of Cuts Through Existing Roadways

- 2.7.1. When jacking below and through road subgrade construction to place pipes or electrical items is not feasible or allowed, cut and remove bituminous pavement or crushed stone road and excavate road subgrade to the minimum extent necessary to properly place the penetrating item.
- 2.7.2. Pipes, ducts, etc. shall be properly bedded, inspected and tested, and backfilled as specified herein, and structural fill (either Type SFG or Type SFC, whichever occurs in construction on both sides of the cut) shall be placed and compacted to a point 250mm below the existing road surface. The Bidder shall compact and prepare material for placement of new bituminous or crushed stone patch by the Bidder .

2.8 Surface Fill of Substation

2.8.1 Materials

- 2.8.1.1. General
Fill material shall be clean, washed gravel or crushed stone and shall be free of foreign matter. Crushed stone shall be tough, durable, free from an excessive quantity of flat, elongated, and soft or disintegrated material.
- 2.8.1.2. Fire Protection Fill
Fire protection fill shall have standard size of #3 gradations in accordance with ASTM D448.
- 2.8.1.3. General Fill
General fill shall conform to the following gradations:

Nominal Size	% by Weight
Square Opening	Passing
38mm	100
25mm	90 ~ 100
13mm	60 ~ 90
No.4	35 ~ 55
No.16	10 ~ 40
No.200	0 ~ 12

2.8.2 Performance

- 2.8.2.1. Preparation of subgrade if required.
- 2.8.2.2. Sod, vegetation, roots, debris and other organic matter shall be removed from the fill areas.
- 2.8.2.3. Subgrade shall be graded as required and shall be free of pockets and high spots and shall be approved by the Owner prior to the placement of fill.
- 2.8.2.4. The fill area shall have special drainage facilities.

2.8.3 Placing of Fill

- 2.8.3.1. Fire protection fill shall be placed in areas designated on the approved Bidder's drawings as "Fire Protection Fill Area". General fill shall be placed in the areas designated on the approved Bidder's drawings as "General Fill Area"
- 2.8.3.2. Fire protection fill and general fill may be placed prior to electrical installation.
- 2.8.3.3. Fire protection fill shall not be rolled, tamped or packed. Finished surface shall be level and no traffic is permitted on fill during placing operations.

- 2.8.3.4. General fill shall be compacted by rolling with a 10 ton or heavier vibratory roller with a minimum of 2 passes per layer. The maximum lift thickness shall be 15cm. Finished surface shall be level.

- 2.8.3.5. The entire substation area within the fence, the building surroundings, along roads, around parking area, trenches and foundations appeared above the ground level shall be covered with a minimum 200 mm thick general surface fill.

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